Assessment 2 Instructions

SOST10142/20142 Applied Statistics for Social Scientists

Please note this assignment should be submitted on-line via Turnitin.

ALL DATA AND RELATED DOCUMENTS FOR THE ASSIGNMENT ARE HELD IN THE 'ASSESSMENT' FOLDER ON THE COURSE BLACKBOARD SITE

The assignment should be submitted as a single document including data analysis report, references, and an appendix with R code. You should use R to produce all materials included in this report (e.g., aesthetically-pleasing figures with appropriate labels, presentation-ready tables, etc.).

The word limit is **maximum 2,500 WORDS (including references).** Please include a word count on the front cover and please note the University policy on plagiarism. The word count *excludes* any titles or text used in tables and graphs, and the appendix.

Step 1. Define the analysis

- Read in the data and take time to study the variables in the dataset. Please consult the codebook and use appropriate functions in R to understand the variable labels. When choosing, take time to understand what each variable is measuring.
- Choose the outcome (dependent) variable for your regression analysis.
- *Your research objective* is to investigate the direction and strength of the relationship between your chosen dependent variable and a set of 'explanatory' variables in the dataset. Choose at least one focal (main) predictor (e.g., var1).
- Then choose at least four 'control' variables (e.g., var2-var5). Your objective is to examine the effect of the focal predictor(s), controlling for these four (or more) variables. You may include additional variables in your model – these are minimum requirements.
- Propose your research question(s). To achieve your objective, it is essential that you specify a clear research question or research objective (and any related testable, directional hypotheses), which will guide your analysis. In selecting your explanatory variables, it is important that you provide a clear reasoning for their inclusion (with some reference to literature – extensive literature reviews are not necessary but some background is necessary to provide context for readers). The selection of variables should be supported by your research objective and appropriate reasoning.

Step 2. Carry out the analysis

1. Provide table of a descriptive statistics.

2. Use appropriate methods (e.g., histograms, boxplots, etc.) to describe the univariate distributions of your dependent and explanatory variables, separately.

3. Produce *two scatterplots* to show the relationship between your dependent variable and each of your explanatory variables, var1 and var2 (so one scatterplot for each dependent-explanatory pair; y-axis is the outcome, while x-axis is the predictor).

NOTE: If you choose a dependent variable and/or independent variable that is measured on a non-continuous scale, it may be helpful to add 'jitter' to your scatterplot(s) to make them more readable. Jitter is random noise around data points and can be added with the `jitter()` function: `plot( jitter(DATA_NAME$VAR1, 3), DATA_NAME$VAR2)` (the '3' tells R how much random noise to add; try different values beginning with 1 and see what happens!).

4. Perform a correlation analysis for all variables to show the linear associations between your dependent and explanatory variables.
5. Conduct a preliminary statistical test using two independent samples t-test. For this, you need to decide a grouping variable, and develop the null and alternative hypothesis. Report the t-test results.
6. Finally, fit at least two linear regression models to investigate the relationship between your dependent variable and your explanatory variables. Model 1 should use the focal(main) predictor (var1) as a single explanatory variable. In model 2, you include the focal variable and remaining control variables, var2-var5.
7. Tidy your R code and prepare to include it in the appendix.

Step 3. Write up the analysis

You should write up using the following report structure (please note the mark scheme and tips on reporting provided below)

Introduction

- Introduce the research question/research objective (and any hypotheses) that you will be investigating with the secondary analysis, with some reference to relevant literature
- Explain the selection of variables used in the analysis, making reference to relevant literature where possible to support your choices.

The Analysis

- Using the table of descriptive statistics, describe the sample size and characteristics. Report any procedures taken (recoding, rescaling, standardizing, for example). The table should include variable labels and categories (if there is a dummy variable, readers need to understand what is 0 and 1). You may consider critique (limitations) of the measures in the dataset, observations (e.g,. ,sampling), etc.
- Visually present your data including scatterplots, correlation analysis and regression (including the relevant outputs from R) with a clear reporting and interpretation of what they show. You may include cross-tabulations (optional).
- Provide your reasoning for conducting a *t*-test using a particular grouping variable. Clarify why you chose the particular grouping variable.
- Analyse regression models by closely examining the changes in the effect of var1. Discuss any shortcomings in the analysis.

Conclusion

- Conclude with a brief summary of your main findings and limitations.

ASSIGNMENT MARKING SCHEME

Introduction to the research including the selection of variables and reporting of the univariate distributions	35%
Reporting/interpretation of main results and conclusion, Presentation of R code	65%

General tips for Report Writing

- Reports benefit from a clear structure. Sensible use of numbered sub-headings will help achieve this.
- You should write concisely to keep to the overall word limit. Note the word limit is 2,500 words with references so you should aim for roughly 2,300 for the report, allocating 200 for references (the exact split is entirely up to you but a good use of referencing can help achieve high marks). Note while it is important to frame your research question with at least some engagement with relevant sources, this essay does NOT require an extensive literature review.
- In reporting your choice of explanatory variables try and give properly reasoned justifications, supported where possible with reference to literature or other sources.
- Avoid simply pasting in tables and graphs without accompanying comment. Readers will want you to draw out the important elements of your output and to explain your results
- Good formulation and presentation of tables and graphs is important. A cut and paste table that goes off the end of the page is a bad table! A consideration on how you present aesthetically pleasing results may help. Remember that for cross-tabulations, appropriate percentages are needed to help you and the reader to interpret the results, so remember to request these when producing your tables.
- Tables and figures included in the report should be numbered and given clear titles, and be referred to in the text.
- Aim to support your analysis with intelligent discussion. For relationships you are reporting, can you offer explanation for the relationship? Did you expect to find it? Might the relationship be influenced by another unexplored variable?
- Focus your discussion of your analysis on the research question being considered – does it support existing research/theory/your own hypotheses?

Specific tips on reporting the analysis for this assignment

- When interpreting correlations, have you considered both the direction (positive or negative) and the strength of the relationship, and whether it is statistically significant.
- In reporting regression results, as well as including the table of results, you should highlight and explain key values in the tables to help with the interpretation.
- You should also express the regression model as an equation and provide a worked example for a scenario.
- When interpreting the regression consider: What will a regression model tell you? Does your explanatory variable have a statistically significant effect on the response variable? Is it substantively large? What is the effect on the response variable of a unit increase in the explanatory variable? How good is the model? How much of the variation in your dependent variable is explained by the model?
- You may consider limitations of the analysis and make comments on them.
- The Steps 1-3 are required to meet the standard of the assessment. You may go beyond this and augment the report, but it is important that you clearly communicate these steps and maintain focus of the report. I will actively look for these! Good luck!